Matrices & Matrix Algebra

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Matrix Multiplication Properties**

Find the standard matrix $[T]$ of the linear operator $T:\mathbb{R}^2\to\mathbb{R}^2$ that performs a horizontal shear by factor 2, followed by a counterclockwise rotation of 45°.

Step 1: Standard matrix of shear $[S]$: $S(x,y)=(x+2y,y)\Rightarrow[S]=(1,2;0,1)$. Step 2: Standard matrix of rotation $[R]$: $[R]=(\cos(45^\circ),\ -\sin(45^\circ);\sin(45^\circ),\cos(45^\circ))=\frac{\sqrt{2}}{2}(1,\ -1;1,1)$. Step 3: Compose operators $[T]=[R][S]$.

2 **[Foundational]** • **Target: Matrix Inverses and Elementary Matrices**

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 3,4\rangle$.

$P=\frac{\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}}=\frac{1}{25}[(9,12;12,16)]=\left[\left(\frac{9}{25},\frac{12}{25};\frac{12}{25},\frac{16}{25}\right)\right]$. Final Answer: $\left[\left(\frac{9}{25},\frac{12}{25};\frac{12}{25},\frac{16}{25}\right)\right]$

3 **[Foundational]** • **Target: Symmetric, Orthogonal, and Special Matrices**

Let $T:\mathbb{R}^2\to\mathbb{R}^2$ have standard matrix $A=(4,1;2,3)$. Find the matrix representation $[T]_{\mathcal{B}}$ relative to the basis $\mathcal{B}=\{(1;1),(-1;2)\}$ via similarity transformation $P^{-1}AP$.

Step 1: Construct change-of-basis transition matrix P:

$P=(1,\ -1;1,2),\quad \det(P)=2-(-1)=3\Rightarrow P^{-1}=\frac{1}{3}(2,1;\ -1,1)$. Step 2: Compute AP:

$AP=(4,1;2,3)(1,\ -1;1,2)=(4(1)+1(1),4(-1)+1(2);2(1)+3(1),2(-1)+3(2))=(5,\ -2;5,4)$. Step 3: Compute $[T]_{\mathcal{B}}=P^{-1}(AP)$.

4 **[Foundational]** • **Target: Linear Transformations and Standard Matrices**

Use the Gauss–Jordan elimination algorithm on the augmented block matrix $[A\mid I_3]$ to compute A^{-1} for: $A=(1,0,2;2,\ -1,3;4,1,8)$.

Step 1: Set up $[A\mid I]$:

$(1,0,2,1,0,0;2,\ -1,3,0,1,0;4,1,8,0,0,1)\to(1,0,2,1,0,0;0,\ -1,\ -1,\ -2,1,0;0,1,0,\ -4,0,1)$. Step 2: Row swap and scaling: $\to(1,0,2,1,0,0;0,1,0,\ -4,0,1;0,\ -1,\ -1,\ -2,1,0)\to(1,0,2,1,0,0;0,1,0,\ -4,0,1;0,0,\ -1,\ -6,1,1)$. Step 3: Clear column 3 to reach reduced row echelon form.

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5 [Foundational] • Target: Matrix Multiplication Properties

Consider a matrix $A = (2, -1; 3, k)$. If the determinant of the matrix A^2 is 100, find the positive value of the parameter k.

The determinant of a matrix $A = (a, b; c, d)$ is given by $\det(A) = ad - bc$. For the given matrix $A = (2, -1; 3, k)$, the determinant is $\det(A) = (2)(k) - (-1)(3) = 2k + 3$. A fundamental property of determinants states that $\det(A^n) = (\det(A))^n$. Therefore, for A^2, we have $\det(A^2) = (\det(A))^2$. We are given that $\det(A^2) = 121$. Substituting the determinant of A, we get $(2k + 3)^2 = 121$. Taking the square root of both sides yields $2k + 3 = \pm\sqrt{121}$, which means.

6 [Foundational] • Target: Matrix Inverses and Elementary Matrices

Prove that if λ is an eigenvalue of an invertible matrix A, then $\lambda eq 0$ and λ^{-1} is an eigenvalue of A^{-1} with the exact same corresponding eigenvector.

Step 1: Show $\lambda eq 0$: By definition of an eigenvalue, there exists a nonzero vector $\mathbf{v} eq \mathbf{0}$ such that $A\mathbf{v} = \lambda\mathbf{v}$. If $\lambda = 0$, then $A\mathbf{v} = \mathbf{0}$, which means $\mathbf{v} \in \text{Null}(A)$, contradicting the assumption that A is invertible. Thus $\lambda eq 0$. Step 2: Show λ^{-1} is an eigenvalue of A^{-1}: Multiply both sides of $A\mathbf{v} = \lambda\mathbf{v}$ by A^{-1} from the left.

7 [Foundational] • Target: Symmetric, Orthogonal, and Special Matrices

Diagonalize the matrix $A = (1, 2; 2, 1)$ by finding invertible P and diagonal D such that $A = PDP^{-1}$. Then compute A^k for any positive integer k.

Step 1: Eigenvalues of A:
$\det(A - \lambda I) = (1 - \lambda)^2 - 4 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 3,\ \lambda_2 = -1.$ Step 2: Eigenvectors:For $\lambda_1 = 3$: $(-2, 2; 2, -2)\mathbf{v}_1 = \mathbf{0} \Rightarrow \mathbf{v}_1 = (1; 1)$.For $\lambda_2 = -1$:
$(2, 2; 2, 2)\mathbf{v}_2 = \mathbf{0} \Rightarrow \mathbf{v}_2 = (-1; 1)$. Step 3: Construct P, D, and P^{-1}.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Linear Transformations and Standard Matrices

For the matrix $A = (1, 2, 0, 1; 2, 4, 1, 4; 3, 6, 1, 5)$, find bases for $\text{Row}(A)$, $\text{Col}(A)$, and $\text{Null}(A)$. Verify the Rank–Nullity Theorem.

Step 1: Row reduction to RREF:
$(1, 2, 0, 1; 2, 4, 1, 4; 3, 6, 1, 5) \to (1, 2, 0, 1; 0, 0, 1, 2; 0, 0, 1, 2) \to (1, 2, 0, 1; 0, 0, 1, 2; 0, 0, 0, 0).$ Step 2: Subspace Bases:Basis for $\text{Row}(A)$: Non-zero rows of RREF: $\{(1, 2, 0, 1), (0, 0, 1, 2)\}$.Basis for $\text{Col}(A)$: Pivot columns of original matrix A (columns 1 and 3): $\{(1; 2; 3), (0; 1; 1)\}$.Basis for $\text{Null}(A)$: Free variables.

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9 [Intermediate] • Target: Matrix Multiplication Properties

Consider a 2×2 matrix A with an eigenvalue $\lambda = 3$ and a corresponding eigenvector $\mathbf{v} = (1; 2)$. If the trace of A is 7, find the other eigenvalue of A.

Let the matrix A be $(a, b; c, d)$. The eigenvalues of A are the roots of the characteristic equation $\det(A - \lambda I) = 0$. We are given that $\lambda_1 = 3$ is an eigenvalue. The trace of a matrix is the sum of its diagonal elements, so $\text{tr}(A) = a + d$. We are given that $\text{tr}(A) = 7$. The trace of a matrix is also equal to the sum of its eigenvalues. Let the eigenvalues be λ_1 and λ_2. Therefore, $\lambda_1 + \lambda_2 = \text{tr}(A)$. Substituting the given.

10 [Intermediate] • Target: Matrix Inverses and Elementary Matrices

Let $A = (1, 2, 0; 0, 3, 1; 2, 1, 4)$. Compute $\det(A)$, find the adjugate matrix $\text{adj}(A)$, and verify the identity $A\text{adj}(A) = \det(A)I_3$.

Step 1: Compute $\det(A)$: Expanding along row 1:
$\det(A) = 1(3(4) - 1(1)) - 2(0(4) - 1(2)) + 0 = 1(11) - 2(-2) = 11 + 4 = 15.$ Step 2: Compute Cofactor Matrix C: $C_{11} = +(12 - 1) = 11, \quad C_{12} = -(0 - 2) = 2, \quad C_{13} = +(0 - 6) = -6$
$C_{21} = -(8 - 0) = -8, \quad C_{22} = +(4 - 0) = 4, \quad C_{23} = -(1 - 4) = 3$
$C_{31} = +(2 - 0) = 2, \quad C_{32} = -(1 - 0) = -1, \quad C_{33} = +(3 - 0) = 3$ The adjugate matrix is the transpose $\text{adj}(A) = C^T$.

11 [Intermediate] • Target: Symmetric, Orthogonal, and Special Matrices

Let A and B be invertible $n \times n$ matrices. Prove algebraically that: (a) $(AB)^T = B^T A^T$, and (b) $(AB)^{-1} = B^{-1}A^{-1}$.

(a) Transpose Proof: The (i, j)-entry of $(AB)^T$ is the (j, i)-entry of AB:
$[(AB)^T]_{ij} = [AB]_{ji} = \sum_{k=1}^{n} A_{jk}B_{ki} = \sum_{k=1}^{n} B_{ki}A_{jk}.$ Using the definition of matrix transpose, $B_{ki} = [B^T]_{ik}$ and $A_{jk} = [A^T]_{kj}$: $[(AB)^T]_{ij} = \sum_{k=1}^{n} [B^T]_{ik}[A^T]_{kj} = [B^T A^T]_{ij}.$ Since entries match for all i, j, $(AB)^T = B^T A^T$. ■ (b) Inverse Proof: A matrix C is the inverse of AB if and only if $(AB)C = I_n$ and $C(AB) = I_n$. Test.

12 [Intermediate] • Target: Linear Transformations and Standard Matrices

Use Cramer's Rule to solve for x_2 in the linear system:
$\{2x_1 + x_2 + x_3 = 4; x_1 - x_2 + 2x_3 = 1; 3x_1 + 2x_2 - x_3 = 7$

Step 1: Compute main determinant $\det(A)$:
$\det(A) = |2, 1, 1; 1, -1, 2; 3, 2, -1| = 2((-1)(-1) - 2(2)) - 1(1(-1) - 2(3)) + 1(1(2) - (-1)(3))$
$= 2(1 - 4) - 1(-1 - 6) + 1(2 + 3) = 2(-3) - 1(-7) + 1(5) = -6 + 7 + 5 = 6.$ Step 2: Replace column 2 with $\mathbf{b} = (4, 1, 7)^T$: $\det(A_2(\mathbf{b})) = |2, 4, 1; 1, 1, 2; 3, 7, -1|.$ Expanding along row 1:
$= 2(1(-1) - 2(7)) - 4(1(-1) - 2(3)) + 1(1(7) - 1(3)) = 2(-15) - 4(-7) + 1(4) = -30 + 28 + 4 = 2.$ Step 3: Evaluate x_2.

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13 [Intermediate] • Target: Matrix Multiplication Properties

Compute the determinant of the 4×4 matrix by Laplace cofactor expansion along an optimal row or column: $A = (2, 0, -1, 3; 0, 5, 0, 0; 1, -2, 3, 1; 4, 0, 1, 2)$.

Step 1: Choose row with maximum zeros: Row 2 has three zero entries: $a_{21} = 0, a_{22} = 5, a_{23} = 0, a_{24} = 0$. Expanding along Row 2: $\det(A) = (-1)^{2+2}(5)|2, -1, 3; 1, 3, 1; 4, 1, 2| = 5|2, -1, 3; 1, 3, 1; 4, 1, 2|$. Step 2: Evaluate 3×3 determinant: $|2, -1, 3; 1, 3, 1; 4, 1, 2| = 2(3(2) - 1(1)) - (-1)(1(2) - 1(4)) + 3(1(1) - 3(4))$ $= 2(6 - 1) + 1(2 - 4) + 3(1 - 12) = 2(5) + 1(-2) + 3(-11) = 10 - 2 - 33 = -25$. Step 3: Multiply by coefficient.

14 [Intermediate] • Target: Matrix Inverses and Elementary Matrices

Consider the matrix $A = (2, 1; 1, 3)$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $(x'; y') = A(x; y)$, what is the image of the point $(3, 1)$ under this transformation?

The transformation is defined by the matrix multiplication $(x'; y') = A(x; y)$. We are given the point $(3, 1)$, so $(x; y) = (3; 1)$ and the matrix $A = (2, 1; 1, 3)$. Performing the matrix multiplication: $(x'; y') = (2, 1; 1, 3)(3; 1) = ((2 \times 3) + (1 \times 1); (1 \times 3) + (3 \times 1)) = (6 + 1; 3 + 3) = (7; 6)$ Therefore, the image of the point $(3, 1)$ is $(7, 6)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Symmetric, Orthogonal, and Special Matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Find the component orthogonal to $\mathbf{u}$ of $\mathbf{y} = \langle -2, 3, -4 \rangle$ for $\mathbf{u} = \langle 1, -3, 3 \rangle$.

$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \frac{-23}{19} \mathbf{u} = \langle -\frac{23}{19}, \frac{69}{19}, -\frac{69}{19} \rangle$, so $\mathbf{y} - \hat{\mathbf{y}} = \langle -\frac{15}{19}, -\frac{12}{19}, -\frac{7}{19} \rangle$. Final Answer: $\langle -\frac{15}{19}, -\frac{12}{19}, -\frac{7}{19} \rangle$

16 [Challenging] • Target: Linear Transformations and Standard Matrices • Cross-Lesson: 4.01 Determinants

Prove that if $\mathbf{x}_p$ is any particular solution to $A\mathbf{x} = \mathbf{b}$, then the complete solution set is the affine subspace $S = \{\mathbf{x}_p + \mathbf{x}_h : \mathbf{x}_h \in \text{Null}(A)\}$.

Two-part set inclusion proof: 1. Show $S \subseteq \text{Solution Set}$: Let $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$ where $A\mathbf{x}_p = \mathbf{b}$ and $A\mathbf{x}_h = \mathbf{0}$. Applying matrix multiplication: $A\mathbf{x} = A(\mathbf{x}_p + \mathbf{x}_h) = A\mathbf{x}_p + A\mathbf{x}_h = \mathbf{b} + \mathbf{0} = \mathbf{b}$. Thus every vector in S is indeed a valid solution. 2. Show $\text{Solution Set} \subseteq S$: Let $\mathbf{y}$ be an arbitrary solution to $A\mathbf{x} = \mathbf{b}$, so.

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17 [Challenging] • Target: Matrix Multiplication Properties • Cross-Lesson: 1.01 Vectors in n Dimensions

Find the shortest Euclidean distance from the point $P(3, -1, 4)$ to the hyperplane $\mathcal{H}: 2x - y + 2z = 5$ in $\mathbb{R}^3$.

Hyperplane Distance Formula: For $\mathcal{H}: \mathbf{n} \cdot \mathbf{x} = d$, the perpendicular distance from $\mathbf{x}_0$ is: $D = \frac{|\mathbf{n} \cdot \mathbf{x}_0 - d|}{||\mathbf{n}||}$. Here $\mathbf{n} = (2, -1, 2)^T$, $d = 5$, and $\mathbf{x}_0 = (3, -1, 4)^T$. 1. Norm of normal vector: $||\mathbf{n}|| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$. 2. Value at point P: $\mathbf{n} \cdot \mathbf{x}_0 = 2(3) - 1(-1) + 2(4) = 6 + 1 + 8 = 15$. 3. Distance.

18 [Challenging] • Target: Matrix Inverses and Elementary Matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Find all eigenvalues, their algebraic and geometric multiplicities, and a basis for each eigenspace for: $A = (3, 1, 0; 0, 3, 0; 0, 0, 2)$. Is A diagonalizable?

Step 1: Characteristic equation: Since A is upper triangular, its eigenvalues are the diagonal entries: $\det(A - \lambda I) = (3 - \lambda)^2(2 - \lambda) = 0$. Distinct eigenvalues: $\lambda_1 = 3$ (algebraic multiplicity $\alpha_1 = 2$), and $\lambda_2 = 2$ (algebraic multiplicity $\alpha_2 = 1$). Step 2: Eigenspace for $\lambda = 3$: $(A - 3I)\mathbf{v} = (0, 1, 0; 0, 0, 0; 0, 0, -1)(v_1; v_2; v_3) = (0; 0; 0)$. We obtain $v_2 = 0$ and $-v_3 = 0 \Rightarrow v_3 = 0$, w.

19 [Challenging] • Target: Symmetric, Orthogonal, and Special Matrices • Cross-Lesson: 4.01 Determinants

Apply the Gram–Schmidt orthogonalization process to the columns of A to find its QR factorization: $A = (1, 2; 2, 1; -2, 2)$.

Step 1: Orthogonalize $\mathbf{x}_1 = (1, 2, -2)^T$ and $\mathbf{x}_2 = (2, 1, 2)^T$: First vector: $\mathbf{v}_1 = \mathbf{x}_1 = (1; 2; -2)$. Norm: $||\mathbf{v}_1|| = \sqrt{1^2 + 2^2 + (-2)^2} = \sqrt{9} = 3$. Subtract projection of $\mathbf{x}_2$ onto $\mathbf{v}_1$: $\mathbf{x}_2 \cdot \mathbf{v}_1 = 2(1) + 1(2) + 2(-2) = 2 + 2 - 4 = 0$. Since $\mathbf{x}_2 \cdot \mathbf{v}_1 = 0$, the columns were already orthogonal! Thus $\mathbf{v}_2 = \mathbf{x}_2 = (2; 1; 2)$. Norm.

20 [Challenging] • Target: Linear Transformations and Standard Matrices • Cross-Lesson: 1.01 Vectors in n Dimensions

Perform a complete bifurcation analysis of the system in terms of parameter $k \in \mathbb{R}$: $\{x + y + kz = 2; 3x + 4y + 2z = k; 2x + 3y - z = 1$ Determine all values of k for which the system has (a) a unique solution, (b) infinitely many solutions, and (c) no solution.

Step 1: Compute determinant of coefficient matrix A: $\det(A) = |1, 1, k; 3, 4, 2; 2, 3, -1|$. Expanding along row 1: $\det(A) = 1(4(-1) - 2(3)) - 1(3(-1) - 2(2)) + k(3(3) - 4(2)) = 1(-10) - 1(-7) + k(1) = k - 3$. Step 2: Classify cases:(a) Unique Solution: If $keq3$, $\det(A)eq0$; the system has full rank 3 and possesses a unique solution.Critical Case $k = 3$: Substitute $k = 3$ into the augmented matrix.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Matrix Multiplication Properties

If matrix A has size 3×4 and matrix B has size 4×2, what is the size of the product AB?

a. 3×2 ✓

b. 4×4

c. 2×3

d. Multiplication is undefined

An $m \times k$ matrix multiplied by a $k \times n$ matrix produces an $m \times n$ matrix. Here $(3 \times 4) \times (4 \times 2) = 3 \times 2$.

22 [Foundational] • Target: Matrix Multiplication Properties

Is matrix multiplication commutative in general ($AB = BA$)?

a. **No, $AB \neq BA$ in general even when both products are defined** ✓

b. Yes, multiplication of numbers and matrices is always commutative

c. Only if the entries are integers

d. Yes, provided both matrices are square

Matrix multiplication is fundamentally non-commutative: in general, $AB \neq BA$.

23 [Foundational] • Target: Matrix Multiplication Properties

What is $(AB)^T$ equal to in terms of the transposes A^T and B^T?

a. $B^T A^T$ ✓

b. $A^T B^T$

c. AB^T

d. BA^T

The transpose of a product reverses order: $(AB)^T = B^T A^T$.

24 [Foundational] • Target: Matrix Inverses and Elementary Matrices

Find the inverse of the 2×2 matrix $A = [[4, 7], [1, 2]]$.

a. $[[2, -7], [-1, 4]]$ ✓

b. $[[-2, 7], [1, -4]]$

c. $[[4, -7], [-1, 2]]$

d. $[[2, 1], [7, 4]]$

$\det(A) = 4(2) - 7(1) = 8 - 7 = 1$. The inverse is $\frac{1}{1}[[2, -7], [-1, 4]] = [[2, -7], [-1, 4]]$.

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25 [Foundational] • Target: Matrix Inverses and Elementary Matrices

What is an elementary matrix?

a. A matrix obtained by performing a single elementary row operation on an identity matrix ✓

b. Any matrix containing only 0s and 1s

c. A diagonal matrix with determinant 1

d. A matrix with exactly one non-zero row

An elementary matrix E represents a single row operation; multiplying EA performs that row operation on A.

26 [Foundational] • Target: Matrix Inverses and Elementary Matrices

If A, B, C are invertible square matrices, what is $(ABC)^{-1}$?

a. $C^{-1}B^{-1}A^{-1}$ ✓

b. $A^{-1}B^{-1}C^{-1}$

c. CBA

d. ABC^{-1}

Inverting a product reverses the sequence of all factors: $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.

27 [Foundational] • Target: Symmetric, Orthogonal, and Special Matrices

A square matrix A is symmetric if and only if:

a. $A^T = A$ ✓

b. $A^T = -A$

c. $A^T = A^{-1}$

d. $\det(A) = 1$

A matrix is symmetric when it equals its own transpose: $A^T = A$ ($a_{ij} = a_{ji}$ for all i, j).

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28 [Foundational] • Target: Symmetric, Orthogonal, and Special Matrices

What defines an orthogonal matrix Q?

a. $Q^TQ = I$, **meaning** $Q^{-1} = Q^T$ ✓

b. $Q^2 = I$

c. $\det(Q) = 0$

d. All entries of Q are perpendicular

An orthogonal matrix has orthonormal columns, which means $Q^TQ = I$ and therefore $Q^{-1} = Q^T$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Symmetric, Orthogonal, and Special Matrices

If Q is an orthogonal matrix, what are the possible values for $\det(Q)$?

a. ± 1 ✓

b. 0

c. 1 only

d. Any real number

$\det(Q^TQ) = \det(I) = 1 \Rightarrow (\det Q)^2 = 1 \Rightarrow \det(Q) = \pm 1$.

30 [Intermediate] • Target: Linear Transformations and Standard Matrices

A transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear if and only if for all vectors $\mathbf{u}$, $\mathbf{v}$ and scalars c:

a. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ **and** $T(c\mathbf{u}) = cT(\mathbf{u})$ ✓

b. $T(\mathbf{u} \cdot \mathbf{v}) = T(\mathbf{u}) \cdot T(\mathbf{v})$

c. $T(\mathbf{u}) = ||\mathbf{u}||$

d. $T(\mathbf{0}) = [1, 1, \ldots, 1]^T$

Linearity is defined by preservation of vector addition and scalar multiplication:
$T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$.

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31 [Intermediate] • Target: Linear Transformations and Standard Matrices

What is the standard matrix for counterclockwise rotation in $\mathbb{R}^2$ by angle θ?

a. $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$ ✓

b. $[[\cos\theta, \sin\theta], [-\sin\theta, \cos\theta]]$

c. $[[\sin\theta, \cos\theta], [\cos\theta, -\sin\theta]]$

d. $[[\cos\theta, 0], [0, \sin\theta]]$

Rotating $[1, 0]^T$ gives $[\cos\theta, \sin\theta]^T$, and rotating $[0, 1]^T$ gives $[-\sin\theta, \cos\theta]^T$, giving $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$.

32 [Intermediate] • Target: Linear Transformations and Standard Matrices

What is the standard matrix for reflection across the x-axis in $\mathbb{R}^2$?

a. $[[1, 0], [0, -1]]$ ✓

b. $[[-1, 0], [0, 1]]$

c. $[[0, 1], [1, 0]]$

d. $[[-1, 0], [0, -1]]$

Reflecting across the x-axis keeps x the same and negates y: $T([x, y]^T) = [x, -y]^T$, so the matrix is $[[1, 0], [0, -1]]$.

33 [Intermediate] • Target: Matrix Multiplication Properties

Consider a linear transformation represented by the matrix $A = (2, 1; 1, -2)$. A point P is mapped to P' by this transformation. If the coordinates of P are $(x; y)$ and the coordinates of P' are $(x'; y')$, then $(x'; y') = A(x; y)$. Suppose a curve is defined by the equation $x'^2 + y'^2 = 5$. What is the equation of the original curve $x^2 + y^2 = k$ in terms of x and y that transforms into this circle? Find the value of k.

a. 1

b. **2** ✓

c. 3

d. 4

The transformation is given by $(x'; y') = (2, 1; 1, -2)(x; y)$. We can express $(x; y)$ in terms of $(x'; y')$ by finding the inverse of matrix A. The determinant of A is $\det(A) = (2)(-2) - (1)(1) = -4 - 1 = -5$. The inverse matrix is $A^{-1} = \frac{1}{-5}(-2, -1; -1, 2) = (2/5, 1/5; 1/5, -2/5)$. Therefore, $(x; y) = A^{-1}(x'; y') = (2/5, 1/5; 1/5, -2/5)(x'; y')$. This giv.

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34 **[Intermediate]** • **Target: Matrix Inverses and Elementary Matrices**

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 1, -1 \rangle$.

a. $[(1, -1; -1, 1)]$

b. $\left[\left(\frac{1}{2}, \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\right)\right]$

c. $[(0, -1; -1, 0)]$

d. $\left[\left(\frac{1}{2}, -\frac{1}{2}; -\frac{1}{2}, \frac{1}{2}\right)\right]$ ✓

$P = \frac{\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}} = \frac{1}{2}[(1, -1; -1, 1)] = \left[\left(\frac{1}{2}, -\frac{1}{2}; -\frac{1}{2}, \frac{1}{2}\right)\right]$.

35 **[Intermediate]** • **Target: Symmetric, Orthogonal, and Special Matrices**

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 2, -1 \rangle$.

a. $[(4, -2; -2, 1)]$

b. $\left[\left(\frac{1}{5}, \frac{2}{5}; \frac{2}{5}, \frac{4}{5}\right)\right]$

c. $\left[\left(\frac{3}{5}, -\frac{4}{5}; -\frac{4}{5}, -\frac{3}{5}\right)\right]$

d. $\left[\left(\frac{4}{5}, -\frac{2}{5}; -\frac{2}{5}, \frac{1}{5}\right)\right]$ ✓

$P = \frac{\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}} = \frac{1}{5}[(4, -2; -2, 1)] = \left[\left(\frac{4}{5}, -\frac{2}{5}; -\frac{2}{5}, \frac{1}{5}\right)\right]$.

36 **[Intermediate]** • **Target: Linear Transformations and Standard Matrices**

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 1, 3 \rangle$.

a. $[(1, 3; 3, 9)]$

b. $\left[\left(\frac{9}{10}, -\frac{3}{10}; -\frac{3}{10}, \frac{1}{10}\right)\right]$

c. $\left[\left(-\frac{4}{5}, \frac{3}{5}; \frac{3}{5}, \frac{4}{5}\right)\right]$

d. $\left[\left(\frac{1}{10}, \frac{3}{10}; \frac{3}{10}, \frac{9}{10}\right)\right]$ ✓

$P = \frac{\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}} = \frac{1}{10}[(1, 3; 3, 9)] = \left[\left(\frac{1}{10}, \frac{3}{10}; \frac{3}{10}, \frac{9}{10}\right)\right]$.

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37 [Intermediate] • Target: Matrix Multiplication Properties

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 1, 1\rangle$.

a. $[(1, 1; 1, 1)]$

b. $\left[\left(\frac{1}{2}, -\frac{1}{2}; -\frac{1}{2}, \frac{1}{2}\right)\right]$

c. $[(0, 1; 1, 0)]$

d. $\left[\left(\frac{1}{2}, \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\right)\right]$ ✓

$P = \frac{\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}} = \frac{1}{2}[(1, 1; 1, 1)] = \left[\left(\frac{1}{2}, \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\right)\right]$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Matrix Inverses and Elementary Matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 1, 2\rangle$.

a. $\left[\left(\frac{1}{5}, \frac{2}{5}; \frac{2}{5}, \frac{4}{5}\right)\right]$ ✓

b. $[(1, 2; 2, 4)]$

c. $\left[\left(\frac{4}{5}, -\frac{2}{5}; -\frac{2}{5}, \frac{1}{5}\right)\right]$

d. $\left[\left(-\frac{3}{5}, \frac{4}{5}; \frac{4}{5}, \frac{3}{5}\right)\right]$

$P = \frac{\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}} = \frac{1}{5}[(1, 2; 2, 4)] = \left[\left(\frac{1}{5}, \frac{2}{5}; \frac{2}{5}, \frac{4}{5}\right)\right]$.

39 [Challenging] • Target: Symmetric, Orthogonal, and Special Matrices • Cross-Lesson: 4.01 Determinants

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 2, 1\rangle$.

a. $[(4, 2; 2, 1)]$

b. $\left[\left(\frac{1}{5}, -\frac{2}{5}; -\frac{2}{5}, \frac{4}{5}\right)\right]$

c. $\left[\left(\frac{4}{5}, \frac{2}{5}; \frac{2}{5}, \frac{1}{5}\right)\right]$ ✓

d. $\left[\left(\frac{3}{5}, \frac{4}{5}; \frac{4}{5}, -\frac{3}{5}\right)\right]$

$P = \frac{\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}} = \frac{1}{5}[(4, 2; 2, 1)] = \left[\left(\frac{4}{5}, \frac{2}{5}; \frac{2}{5}, \frac{1}{5}\right)\right]$.

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40 [Challenging] • Target: Linear Transformations and Standard Matrices • Cross-Lesson: 1.01 Vectors in n Dimensions

Consider the matrix $A = (k, 2; 1, -3)$. If the matrix A is invertible and its inverse A^{-1} has a determinant of $-\frac{1}{11}$, find the value of k.

a. **4** ✓

b. 5

c. 3

d. 6

The determinant of matrix A is given by $\det(A) = (k)(-3) - (2)(1) = -3k - 2$. For an invertible matrix, its determinant is non-zero. The determinant of the inverse of a matrix is the reciprocal of the determinant of the original matrix, i.e., $\det(A^{-1}) = \frac{1}{\det(A)}$. We are given that $\det(A^{-1}) = -\frac{1}{11}$. Therefore, $-\frac{1}{11} = \frac{1}{-3k-2}$. This implies $-11 = -3k - 2$. Rearranging the equation to solve for k: $-11 + 2 = -3k$ $-9 = -3k$ $k = \frac{-9}{-3} = 3$. However, let's.

41 [Challenging] • Target: Matrix Multiplication Properties • Cross-Lesson: 2.01 Systems of Linear Equations

Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (3, -1\ 1, 2)$ with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. Consider a new basis $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 = (1\ 1)$ and $\mathbf{v}_2 = (1\ -1)$. Let P be the change-of-basis matrix from basis B to the standard basis. Find the matrix representation of T with respect to basis B, denoted by $[T]_B$. This matrix can be expressed as $P^{-1}AP$. Find the sum of the diagonal entries of $[T]_B$.

a. **5** ✓

b. 6

c. 4

d. 7

The change-of-basis matrix P from basis B to the standard basis has the vectors of B as its columns. Thus, $P = (1, 1\ 1, -1)$. The inverse of P is $P^{-1} = \frac{1}{\det(P)}(-1, -1\ -1, 1)$. Since $\det(P) = (1)(-1) - (1)(1) = -2$, we have $P^{-1} = \frac{1}{-2}(-1, -1\ -1, 1) = (1/2, 1/2\ 1/2, -1/2)$. The matrix representation of T with respect to basis B is given by $[T]_B = P^{-1}AP$. First, we compute AP.

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42 [Challenging] • Target: Matrix Inverses and Elementary Matrices • Cross-Lesson: 4.01 Determinants

Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(x, y) = (3x - y, x + 2y)$. Consider the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for $\mathbb{R}^2$, where $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$. Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ be another basis for $\mathbb{R}^2$, where $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -1)$. If the matrix representation of T with respect to the basis $\mathcal{B}$ is $[T]_{\mathcal{B}} = (a, b\ c, d)$, and $a + d = 5$, determine the value of $\det(T)$.

a. **7 ✓**

b. 5

c. 3

d. 2

The matrix representation of T with respect to the standard basis $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$ is $A = (3, -1\ 1, 2)$. The determinant of T is the determinant of its matrix representation with respect to any basis, so $\det(T) = \det(A) = (3)(2) - (-1)(1) = 6 + 1 = 7$. The information about the basis $\mathcal{B}$ and $[T]_{\mathcal{B}} = (a, b\ c, d)$ with $a + d = 5$ is a distractor. The trace of a matrix is invariant under a change of basis, so.

43 [Challenging] • Target: Symmetric, Orthogonal, and Special Matrices • Cross-Lesson: 1.01 Vectors in n Dimensions

Let A be a 2×2 matrix given by $A = (2, k; 3, -1)$. If $A^2 = (1, -7; 3, 4)$, find the value of k.

a. 2

b. **3 ✓**

c. 4

d. 5

We are given the matrix $A = (1, 2; k, 3)$. The transpose of A, denoted by A^T, is obtained by interchanging its rows and columns. Thus, $A^T = (1, k; 2, 3)$ We are given the equation $A + A^T = (2, 5; 5, 6)$. Let's compute the sum $A + A^T$:
$A + A^T = (1, 2; k, 3) + (1, k; 2, 3) = (1 + 1, 2 + k; k + 2, 3 + 3) = (2, 2 + k; k + 2, 6)$ Now, we equate this resulting matrix to the given matrix $A + A^T = (2, 5; 5, 6)$.

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44 [Challenging] • Target: Linear Transformations and Standard Matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (2, k\ 1, 3)$ with respect to the standard basis. The image of the vector $\mathbf{v} = (1\ 1)$ under T is $T(\mathbf{v}) = (3\ 4)$. Find the value of the parameter k.

a. **1** ✓

b. 2

c. 3

d. 4

The linear transformation T is represented by the matrix $A = (2, k\ 1, 3)$. The action of T on a vector $\mathbf{v}$ is given by $T(\mathbf{v}) = A\mathbf{v}$. We are given that $\mathbf{v} = (1\ 1)$ and $T(\mathbf{v}) = (3\ 4)$. Therefore, we have the matrix equation: $(2, k\ 1, 3)(1\ 1) = (3\ 4)$ Performing the matrix multiplication on the left side gives: $(2(1) + k(1)\ 1(1) + 3(1)) = (2 + k\ 4)$ Equating this to.

45 [Challenging] • Target: Matrix Multiplication Properties • Cross-Lesson: 4.01 Determinants

A linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(x, y) = (ax + by, cx + dy)$. The matrix representation of T with respect to the standard basis is $A = (a, b\ c, d)$. The image of the standard basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ under T are given by $T(\mathbf{e}_1) = (3, -1)$ and $T(\mathbf{e}_2) = (1, 2)$. The transformation is applied to a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$. The area of the transformed square is known to be k. Find the value of k.

a. **5** ✓

b. 10

c. 15

d. 20

The matrix representation of the linear transformation T with respect to the standard basis is formed by the images of the standard basis vectors as its columns. Therefore, $A = (3, 1\ -1, 2)$. The area of the transformed region is the absolute value of the determinant of the matrix A multiplied by the area of the original region. The original region is a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$, which has an area of $1 \times 1 = 1$. The determinant of A is.

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